BCS-051	Principles of Programming Language	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	To introduce the major programming paradigms, and the principles and techniques involved in design and implementation of modern programming languages
CO2	To introduce notations to describe syntax and semantics of programming languages.
CO3	To analyze and explain behavior of simple programs in imperative languages using concepts such as binding, scope, control structures, subprograms and parameter passing mechanisms.
CO4	To introduce the concepts of ADT and object oriented programming for large scale software development.
CO5	To introduce the concepts of concurrency control and exception handling

Detailed Syllabus

MODULE-I

Introduction: Evolution of language designs, Evaluation criteria, Programming environments, Issues in language translation- Syntactic and semantic rules of a Programming language, Stages in Translation. Characteristics of a good programming language, Programming language translators- compiler & interpreters. Programming Languages: Introduction to Procedural, non-procedural, structured, functional and object oriented programming language, Comparison of C & C++ programming languages.

MODULE-II

Data Type: Declarations, Assignment & initialization, Elementary data types- integer, floating point and fixed point real numbers, character, Boolean, Other numeric data types. Character string, User defined- Enumeration and subrange, Array- one and two dimensional, Records- multilevel and variant, Pointer and reference types. Names, Variables, Concept of binding and types, Type checking, Strong typing, Type compatibility, sequence control with expressions, Conditional statements, Loops.

MODULE-III

Subprograms: Fundamental of subprograms, Scope and life time of variable, Static and dynamic scope, Design issues of subprogram, Local referencing environment, Parameter passing methods, Overloaded subprograms, Generic subprograms, Coroutines. Abstract data types- Abstraction and encapsulation

Text Books:

1. Programming languages Design & implementation by T.W.Pratt, 1996, PHI.
2. Programming Languages – Principles and Paradigms by Allen Tucker & Robert Noonan, 2002, TMH.

Reference Books :

1. Fundamentals of Programming languages by Ellis Horowitz, 1984, Galgotia Publications
2. Programming languages concepts by C. Ghezzi, 1989, Wiley Publications.
3. Programming Languages–Principles and Pradigms Allen Tucker , Robert Noonan 2002, T.M.H

Course Outcomes: After the completion of the course the student will be able to:

CO1	Knowledge of, and ability to use, language features used in current programming languages.
CO2	Aware about the basics concept of programming language
CO3	An ability to program in different language paradigms and evaluate their relative benefits.
CO4	An understanding of the key concepts in the implementation of common features of programming languages
CO5	Understand the basics concept of object oriented programming